

Summary

MERN Stack Engineer with 3+ years building high-scale, distributed backend systems — Node.js, event-driven pipelines with Bull queues, multi-layer Redis caching, and real-time communication via WebSockets. Experience designing low-latency, horizontally scalable services across edtech and healthtech products.

Experience

Software Engineer, Feb 2024 – Present

Code Brew Labs, Chandigarh

- Architected and developed the backend for an **E-learning platform** using Node.js, MongoDB, Redis, and AWS S3, **exposing 50+ RESTful APIs** across authentication, course management, payments, and admin modules with role-based access control.
- Integrated **Razorpay payment gateway** supporting order creation, signature verification, automated refunds, promo codes, and loyalty points for course, package, and event purchases.
- Implemented **multi-layer distributed caching** (in-memory + Redis) with Pub/Sub invalidation, reducing API latency by **60%** under high-concurrency traffic.
- Designed **asynchronous, event-driven pipelines using Bull queue** with retry, backoff, and dead-letter handling, improving throughput and fault tolerance under load.
- Set up **CI/CD pipelines using GitHub Actions** and wrote **unit and integration tests with Jest**, automating builds and test runs to catch regressions before deployment.
- Built **frontend modules in React** with Context API and memoization, integrating REST APIs via Axios with a reusable component architecture for course management and user dashboards.
- Built secure **S3 multipart upload pipeline** using presigned URLs and STS tokens, enabling resumable uploads of multi-GB course videos across unstable networks.
- Developed backend services for the **Portl Fitness app (Shark Tank featured)** using **Node.js, MongoDB, and TypeScript**, including authentication, workout analytics, and admin reporting.
- Developed **analytics services using MongoDB aggregation pipelines** to generate weekly, monthly, and yearly insights for administrative dashboards.

Software Engineer, June 2023 – Jan 2024

UpfrontHealthCare, Mohali

- Optimized high-latency **SQL reporting queries** by implementing **indexing, partitioning, and execution plan analysis**, significantly improving data retrieval performance.
- Developed an automated **data archiving system using MySQL stored procedures** with partitioning strategies to reduce load on the primary production database.
- Built **low-latency real-time communication** systems using **WebSockets and Server-Sent Events (SSE)**, handling concurrent connections efficiently.
- Integrated and managed healthcare data pipelines using **Azure Data Factory**, enabling efficient processing and transformation of large datasets.
- Implemented **data cleansing and transformation workflows** using Azure Dataflows to remove duplicates and improve data quality before storage.

Achievements

- Led end-to-end implementation of the **complete payment flow** for the edtech platform, covering order creation, Razorpay integration, automated refunds, promo codes, and GST e-invoice generation.
- **Mentored 4 interns** at Code Brew Labs — conducted regular code reviews, enforced best practices, and guided them on backend development with Node.js and MongoDB.
- **LeetCode (chirag_0401):** Solved **400+ problems**; earned Top SQL 50 and 100 Days badges — strong in Recursion, Backtracking, and Trees.

Technologies

- **Backend:** Node.js, Express.js, TypeScript, JavaScript
- **Frontend:** React, Context API, Axios
- **Databases:** MySQL, MongoDB, PostgreSQL, Redis
- **Cloud & DevOps:** AWS (S3, Multipart), Docker, PM2
- **Concepts:** Distributed systems, Event-driven architecture, Horizontal scaling, Caching, WebSockets, Rate limiting

Education

B.Tech, Computer Science
Chitkara University

Feb 2019 – June 2023
CGPA: 9.4 / 10